



**College Of Technology
Kandy**

**NVQ Level 5 National Diploma In Information &
Communication Technology**

Sprinkle Dreams Billing System Report

Name of Student	: P. SAJANI SANDEEPANI
Registration Number	: KY/23/ETB11/1/0007
Subject	: SOFTWARE PROGRAMMING
Batch	: 2023/2024

Project Title : Sprinkle Dreams Billing System

Purpose : Cake Ordering

Date :/...../2025

Prepared by : P.Sajani Sandeepani

Approved by :

Name

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Signature

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1. Introduction

Sprinkle Dreams is a small bakery famous for its custom cakes and desserts. As more customers have started coming in, it has become clear that a dependable and automated billing system is needed. This project focuses on creating an easy-to-use and efficient billing system. It will simplify customer transactions, keep track of inventory, and produce financial reports.

2. Purpose of the Project

The primary goal of the billing system is to automate manual billing processes, reduce human errors, and increase efficiency. It will enable the shop to:

- Create and manage customer bills.
- Track inventory usage and restocking.
- Support decision-making through data analytics.

3. Requirement Specification

Hardware Requirement Specification:

- Processor : Intel(R) Core(TM) i5-6200U CPU @ 2.30GHz, 2401 Mhz, 2 Core(s), 4 Logical Processor(s)
- RAM : 4GB
- System Name : DESKTOP-1KTQ262

Software Requirement Specification:

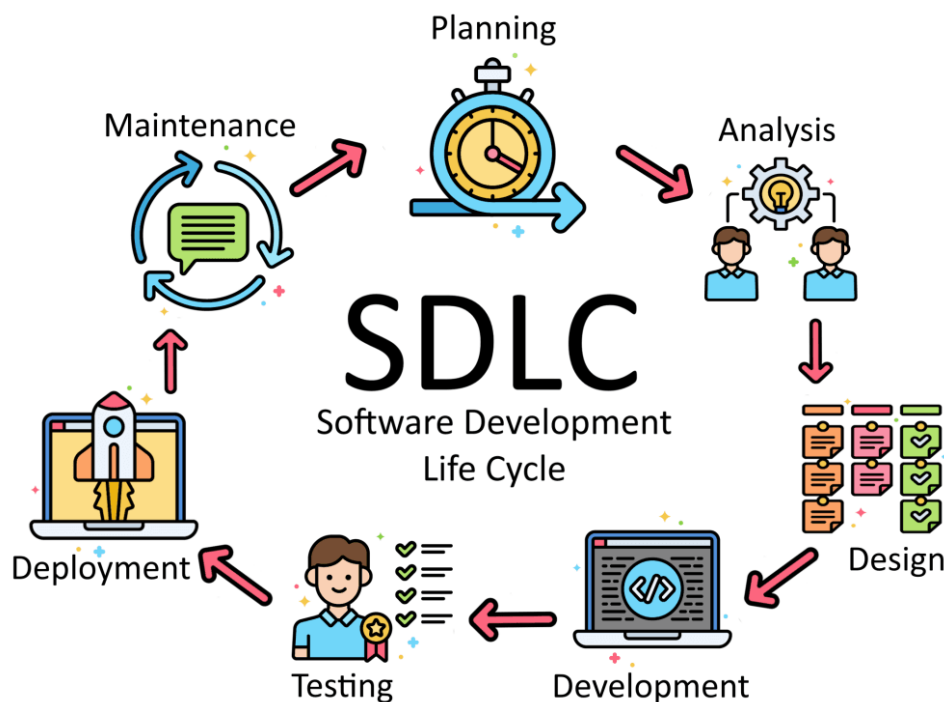
- OS Name : Microsoft Windows 10 Home Single Language
- Programming Name : Java Swing
- IDE : Net beans IDE 25
- Database : Mysql 8.0.31

4. System Development Life Cycle

The System development life cycle (SDLC) is the process of determining how an information system (IS) can support business needs, designing the system, building it and delivering it to users. If you have taken a programmed on your own, this probably sounds pretty simple. In the real world, however, it is not easy.

Phases of Software Development Life Cycle (SDLC)

- Planning
- Analysis
- Design
- Development
- Testing
- Deployment
- Maintenance



Planning

Feasibility Report

Economic Feasibility

- Low-cost development using open-source tools.
- Return on investment through reduced manual labor and errors.

Technical Feasibility

- Technology used is well-supported and suitable for small-scale business operations.

Operational Feasibility

- Daily operations will become faster and more accurate.

Gathering Requirements & Analysis

Gathering Requirements

Stakeholders

- Shop Owner
- Customers
- System Administrator

Methods Used

- Interviews with the shop owner. (My Thoughtlets)
- Observation of current billing process.

Identified Needs

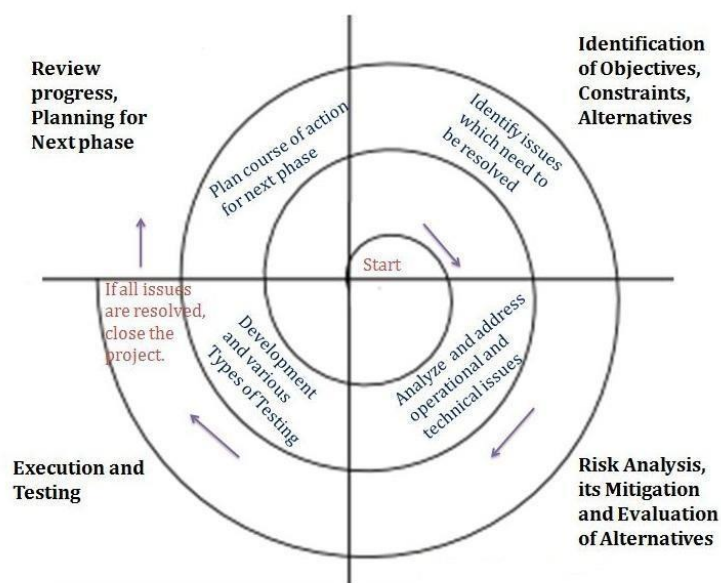
- Fast invoice generation
- Category & product management
- Discounts calculations

Functional Requirements

Functional Requirement	Non-Functional Requirement
• Users can log in (use only me)	• System should be user-friendly and intuitive
• Add/Edit/Delete cakes and categories	• Response time must be less than 2 seconds
• Capture customer details (Name, Contact Number, Email etc)	• Data must be stored securely in a relational database
• Create and save orders with multiple items	• Compatibility with standard printers for invoice printing
• Automatically calculate total amount, discount	
• Generate printable invoices	
• View past sales reports	
• Search orders by date or customer	

Analysis

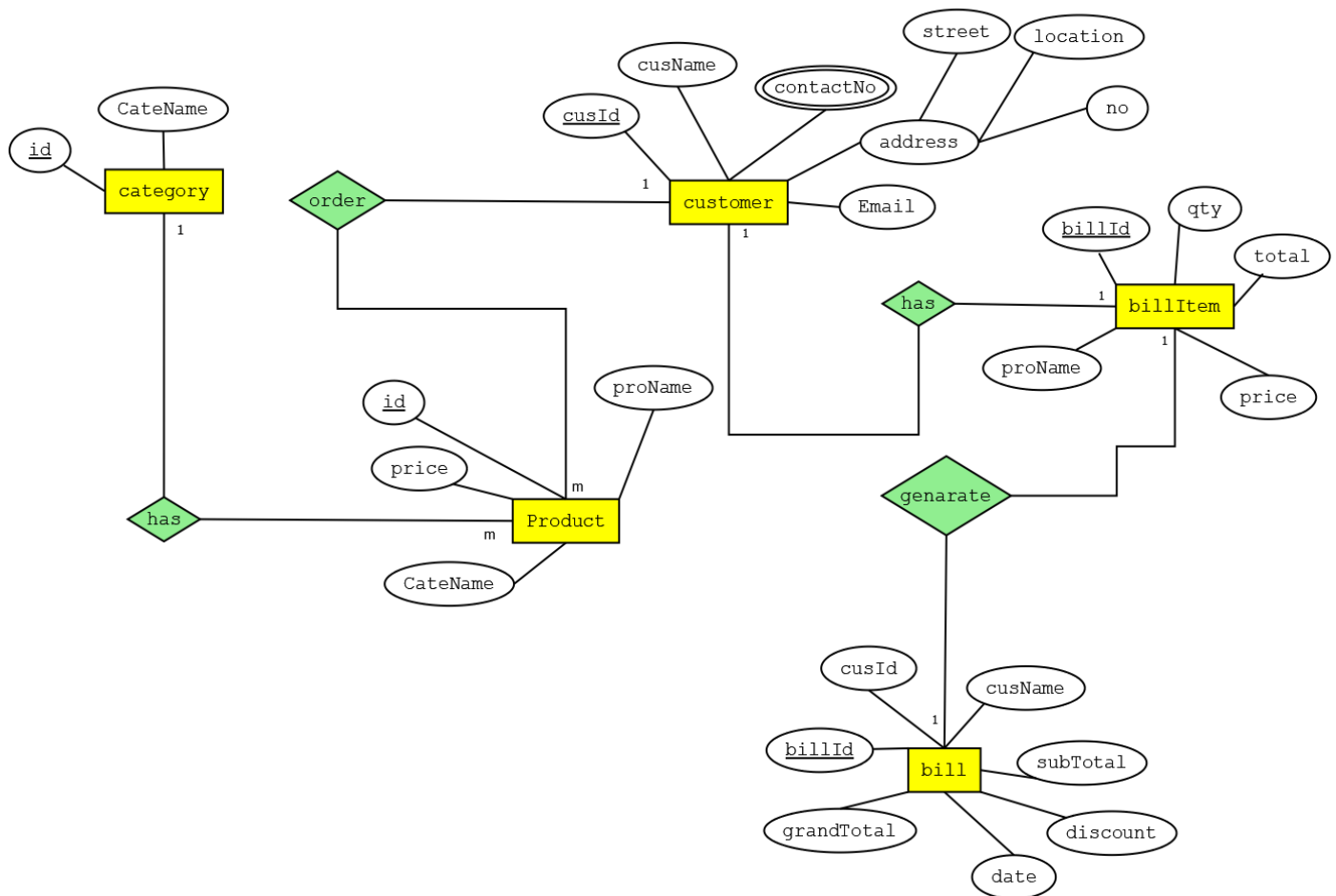
This document plays a vital role in development of life cycle as it describes the complete requirements of the system. **Spiral Model** was defined by Barry Boehm in his 1998 article “A spiral models of Software Development and Enhancement”. This model was not the first model to discuss iterative development, but it was the first model to explain why the iteration models. Spiral model is a combination of sequential and prototype model. This model is best used for large projects which involves continuous enhancements. Where the output is a small prototype of the large software.



Design

ER Diagram

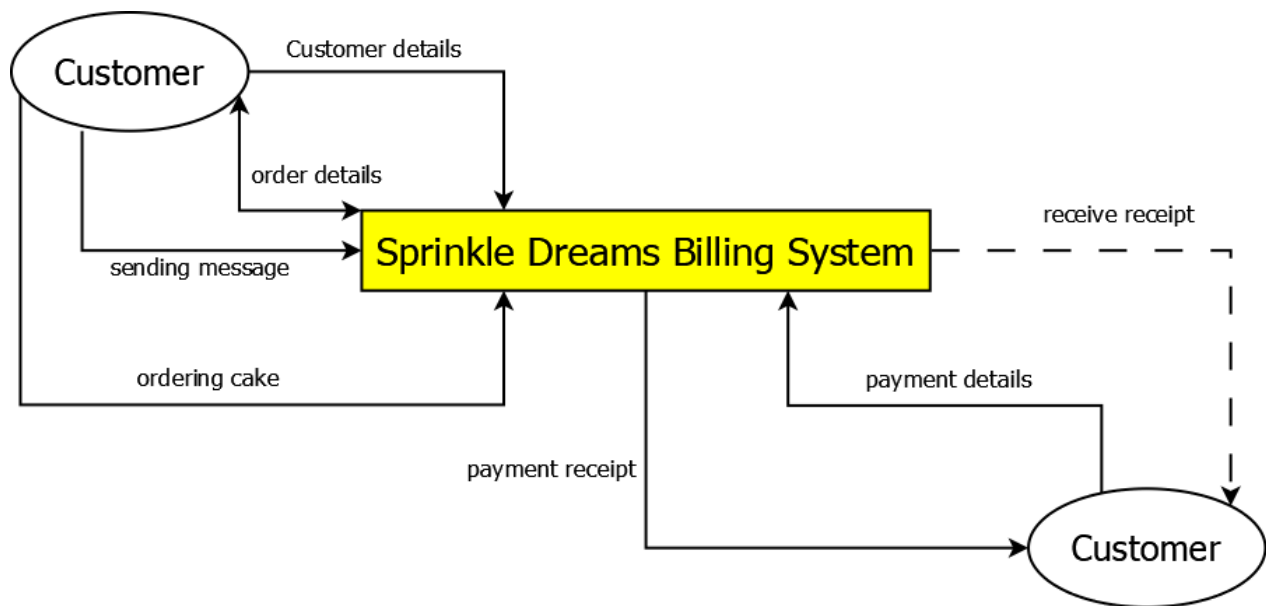
Sprinkle Dreams Billing System



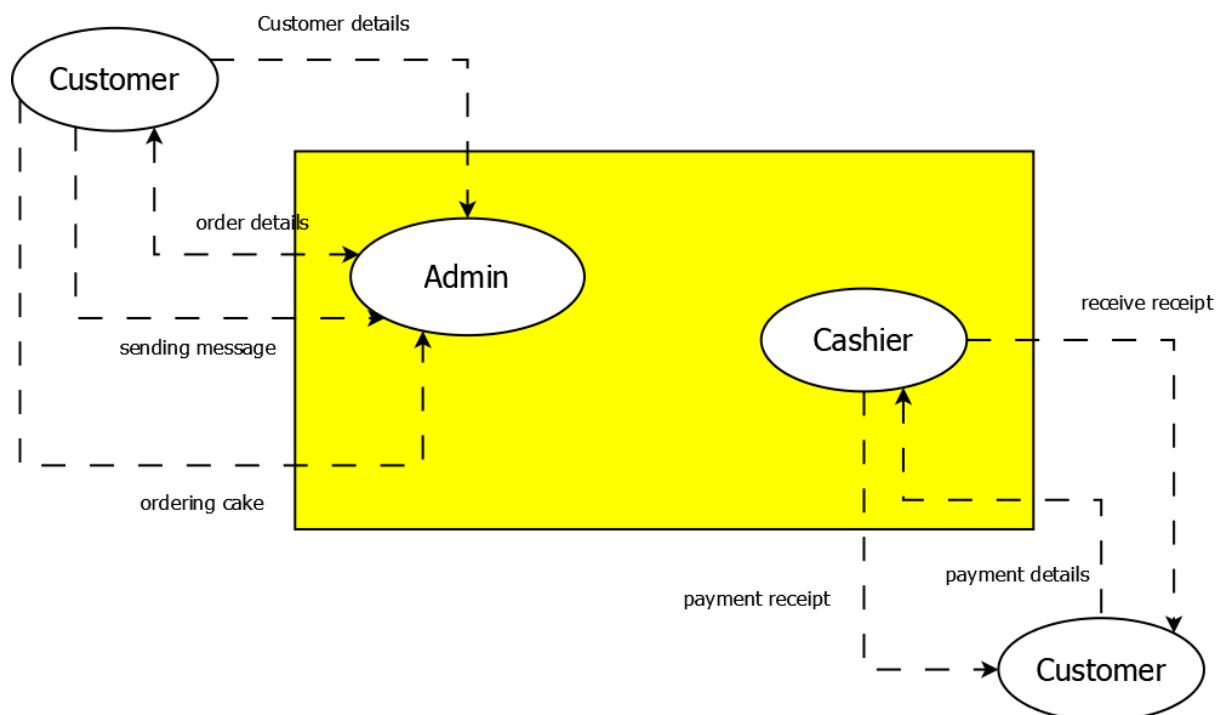
Relationship

- Category has product – one to many
- Customer Order Product – one to many
- Customer has Bill Item – One to One
- Bill Item Generate Bill – One to One

Context Diagram



Document Flow Diagram (DFD)



Document Flow Diagram (DFD)

Development

Development Stages:

- Database schema creation
- User interface forms
- Core functionalities

Testing

- **Unit Testing** : Each module tested individually (e.g., login, adding products)
- **Integration Testing** : End-to-end workflows tested (placing orders, generating bills)
- **System Testing** : Shop owner verified all scenarios work as expected

Deployment

- Application packaged into an executable .jar file
- Database hosted on local server
- Deployed to cashier computer
- Printer configured for invoices

Maintenance

- Regular backups scheduled weekly
- Bug fixes as needed
- Option to add new features (e.g., SMS notifications)
- Database cleanup every 6 months

5. Database Setup

Create the database - Set up the tables for Users, Categories, Product, Customer, Bill item and Bill.

Table	Action	Rows	Type	Collation	Size	Overhead
☐ bill	★ Browse Structure Search Insert Empty Drop	84	InnoDB	utf8mb4_0900_ai_ci	16.0 KiB	-
☐ bill_item	★ Browse Structure Search Insert Empty Drop	122	InnoDB	utf8mb4_0900_ai_ci	16.0 KiB	-
☐ categories	★ Browse Structure Search Insert Empty Drop	7	InnoDB	utf8mb4_0900_ai_ci	16.0 KiB	-
☐ customer	★ Browse Structure Search Insert Empty Drop	11	InnoDB	utf8mb4_0900_ai_ci	16.0 KiB	-
☐ product	★ Browse Structure Search Insert Empty Drop	20	InnoDB	utf8mb4_0900_ai_ci	16.0 KiB	-
☐ users	★ Browse Structure Search Insert Empty Drop	4	InnoDB	utf8mb4_0900_ai_ci	16.0 KiB	-
6 tables	Sum	248	InnoDB	utf8mb4_0900_ai_ci	96.0 KiB	0 B

6. Database Table Structure

The database contains the following tables.

➤ Users

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐ 1	id 🔑	int			No	None		AUTO_INCREMENT	Change Drop More
☐ 2	Name	varchar(100)	utf8mb4_0900_ai_ci		Yes	NULL			Change Drop More
☐ 3	Email	varchar(100)	utf8mb4_0900_ai_ci		Yes	NULL			Change Drop More
☐ 4	Username	varchar(100)	utf8mb4_0900_ai_ci		Yes	NULL			Change Drop More
☐ 5	Password	varchar(100)	utf8mb4_0900_ai_ci		Yes	NULL			Change Drop More

➤ Categories

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐ 1	id 🔑	int			No	None		AUTO_INCREMENT	Change Drop More
☐ 2	name	varchar(50)	utf8mb4_0900_ai_ci		Yes	NULL			Change Drop More

➤ Product

	#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1	id	int			No	None		AUTO_INCREMENT	Change Drop More
☐	2	name	varchar(100)	utf8mb4_0900_ai_ci		Yes	NULL			Change Drop More
☐	3	category	varchar(50)	utf8mb4_0900_ai_ci		Yes	NULL			Change Drop More
☐	4	price	double			Yes	NULL			Change Drop More

➤ Customer

	#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1	id	int			No	None		AUTO_INCREMENT	Change Drop More
☐	2	name	varchar(100)	utf8mb4_0900_ai_ci		Yes	NULL			Change Drop More
☐	3	mobile	varchar(15)	utf8mb4_0900_ai_ci		Yes	NULL			Change Drop More
☐	4	email	varchar(100)	utf8mb4_0900_ai_ci		Yes	NULL			Change Drop More
☐	5	address	text	utf8mb4_0900_ai_ci		Yes	NULL			Change Drop More

➤ Bill item

	#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1	id	int			No	None		AUTO_INCREMENT	Change Drop More
☐	2	bill_id	int			Yes	NULL			Change Drop More
☐	3	product_name	varchar(100)	utf8mb4_0900_ai_ci		Yes	NULL			Change Drop More
☐	4	price	double			Yes	NULL			Change Drop More
☐	5	quantity	int			Yes	NULL			Change Drop More
☐	6	total	double			Yes	NULL			Change Drop More

➤ Bill

	#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1	id	int			No	None		AUTO_INCREMENT	Change Drop More
☐	2	customer_id	int			Yes	NULL			Change Drop More
☐	3	date	timestamp			Yes	CURRENT_TIMESTAMP		DEFAULT_GENERATED	Change Drop More
☐	4	subtotal	double			Yes	NULL			Change Drop More
☐	5	discount	double			Yes	NULL			Change Drop More
☐	6	grandtotal	double			Yes	NULL			Change Drop More

7. Output Screens

➤ Signup

 SIGNUP





Sprinkle Dreams

Baking Delightful Memories....
Cake will love you forever....

SIGN UP

Name

Email

Username

Password

Sign Up

I've an account **Login**

➤ Login

 LOGIN





Sprinkle Dreams

Baking Delightful Memories....
Cake will love you forever....

LOGIN

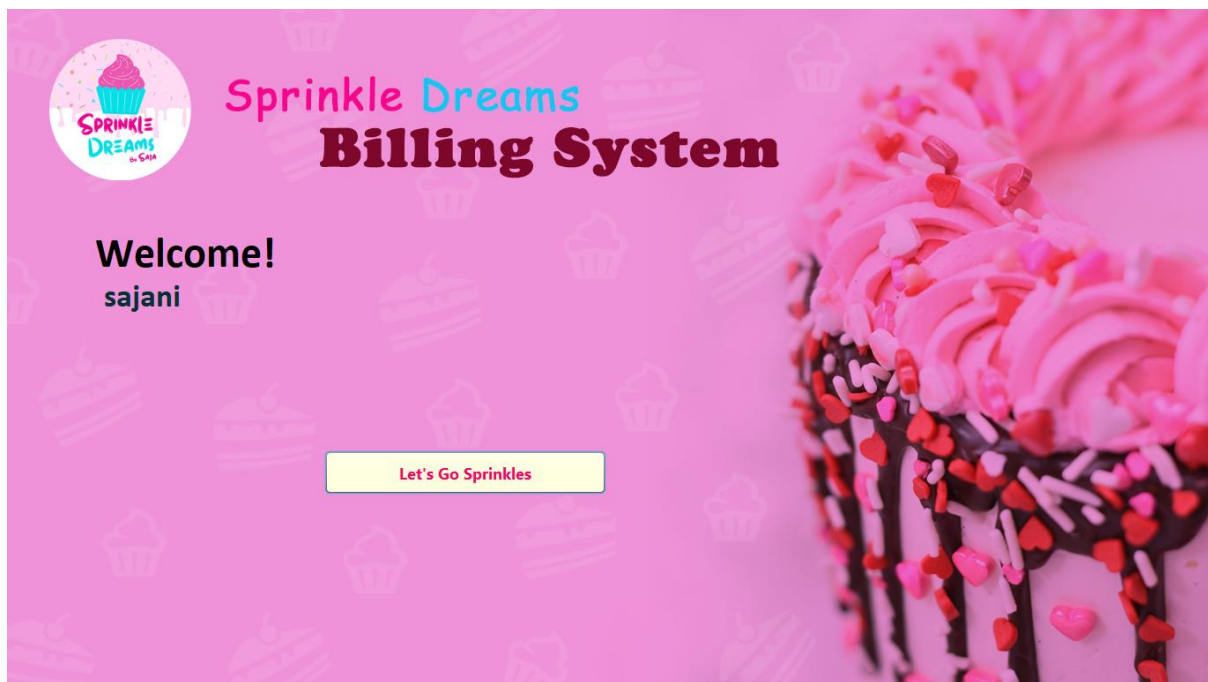
Uername

Password

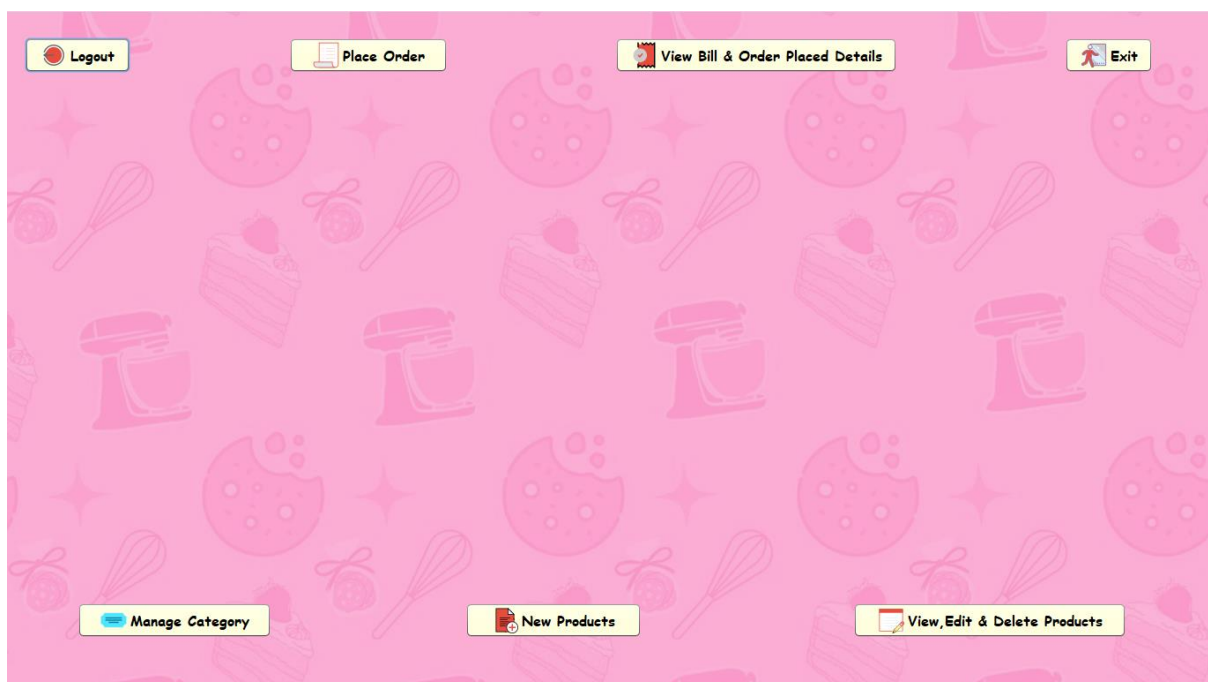
Login

I don't have an account **Sing Up**

➤ Home



➤ Dashboard



➤ **Place Order**

[illegible]

➤ **View Bills & Order Placed Details**



View Bills & Order Placed Details

Date

Order ID

Bill ID	Customer Name	Date	Grand Total
1	Preme	2025-06-27 23:26:14	5700.0
2	Preme	2025-06-27 23:26:20	5700.0
3	Sachi	2025-06-27 23:36:56	2850.0
4	Sachi	2025-06-27 23:36:56	2850.0
31	Sajani Sandeepani	2025-06-28 00:15:39	4294.0
32	Sajani Sandeepani	2025-06-28 00:15:39	4294.0
75	Imesh	2025-06-28 02:16:21	4275.0
76	Imesh	2025-06-28 02:16:21	4275.0
77	Prasangika Madhushani	2025-06-28 11:11:59	7144.0
78	Prasangika Madhushani	2025-06-28 11:11:59	7144.0
79	Rashmi Keshani	2025-07-02 21:03:46	4275.0
80	Rashmi Keshani	2025-07-02 21:03:46	4275.0
81	Sathsarani	2025-07-07 01:47:06	2622.0
82	Sathsarani	2025-07-07 01:47:07	2622.0
83	Malkanthi	2025-07-12 19:00:06	3382.0
84	Malkanthi	2025-07-12 19:00:06	3382.0

➤ **Manage Category**

 **Manage Category**

Add New Category

 Save

 Clear

View Category

ID	Category
1	Cakes
2	Birthday Cakes
3	Mini cakes
4	Cup Cakes & Jar Cakes
5	Brownies
6	Chocolate Items
7	Sweet

Clicked on row to Delete Category

➤ **New Product**

 **New Product**

Name

Category

Cakes

Price

 Save

 Clear

➤ View, Edit & Delete Products

ID

00

Name

Category

Cakes

Price

Update

Delete

Clear

ID	Name	Category	Price
1	Butter Cake	Cakes	1800.0
3	Chocolate Cake	Cakes	2200.0
4	Ribbon Cake	Cakes	2100.0
5	Marble Cake	Cakes	2200.0
6	Velvet Cake	Cakes	2500.0
7	Coconut Cake	Cakes	2500.0
8	Butter Cake with Icing	Cakes	2000.0
9	Chocolate Cake	Cakes	2200.0
10	Benthoo Cake	Mini cakes	2500.0
11	Benthoo Cake with Cup...	Mini cakes	3000.0
12	Fruit Cake	Cakes	2000.0
13	Butter cup cake	Cup Cakes & Jar Cakes	60.0
14	Chocolate cup cake	Cup Cakes & Jar Cakes	80.0
15	Butter cup cake with ic...	Cup Cakes & Jar Cakes	160.0
16	Chocolate cup cake wi...	Cup Cakes & Jar Cakes	160.0
17	Jar Cake	Cup Cakes & Jar Cakes	320.0
18	Fudgy Brownies	Brownies	170.0
20	Mini Chocolate Cake c...	Chocolate Items	220.0
21	Mini Heart Cake Bites	Chocolate Items	240.0
22	Vanila Cake	Birthday Cakes	4500.0

➤ Invoice

***** SPRINKLE DREAMS *****

C28, Ambagamuwa, Udabulathgama

071-1675949

INVOICE

Bill ID : SD86

Customer : Sajani Sandeepani

Mobile : 0712698776

Date/Time : 2025-07-13 22:25:54

Item Name	Price	Qty	Total
Butter cup cake with Icing	160.0	5	800.0
Vanila Cake	4500.0	1	4500.0
Sub Total :			5300.00
Discount (5%) :			265.00
Grand Total :			5035.00

THANK YOU

For being an amazing customer!

And Having Fantastic Taste...

Come again!

8. Test Cases for Sprinkle Dreams Billing System

➤ User Login

Test Case No	Description	Steps	Test Data	Expected Result	Status
TC001	Valid Login	a. Open login form b. Enter valid username and password c. Click "Login"	Username: "saja" Password: "123"	Login successful, home page displayed	Pass
TC002	Invalid Login - Incorrect Password	a. Open login form b. Enter valid username and incorrect password c. Click "Login"	Username: "saja" Password: "1234"	Error message: "Incorrect Password"	Pass
TC003	Invalid Login - Incorrect Username	a. Open login form b. Enter invalid username and correct password c. Click "Login"	Username: "sajani" Password: "123"	Error message: "Username not found "	Pass

➤ Manage Category

Test Case No	Description	Steps	Test Data	Expected Result	Status
TC004	Delete category	a. Open manage category form b. Click "category Row"	Delete Confirmation – “Do You want to delete this category?” Yes/No	“Category deleted Successfully!”	Pass
TC005	Add category	a. Open manage category form b. Click "save"	ok	“Category added Successfully!”	Pass

➤ **New Product**

Test Case No	Description	Steps	Test Data	Expected Result	Status
TC006	Add product	c. Open new product form d. Click "save"	ok	"Product saved added Successfully!"	Pass

➤ **View, Edit & Delete Products**

Test Case No	Description	Steps	Test Data	Expected Result	Status
TC007	Update product	a. Open view, edit & delete Products form b. Click "Update"	ok	"Product Updated"	Pass
TC008	Delete product	c. Open view, edit & delete Products form d. Click "Delete"	ok	"Product Deleted"	Pass

9. Conclusion

The Sprinkle Dreams Cake Billing System automates billing, improves accuracy, and saves time. It is user-friendly, reliable, and ready to support the shop's daily operations effectively.